MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, TIME DELAY (ON RELEASE), CLASS B, TYPE IIA, HERMETICALLY SEALED, 4PDT, 10 AMPERES, FIXED TIME DELAY, 0.1 TO 600 SECONDS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83726.

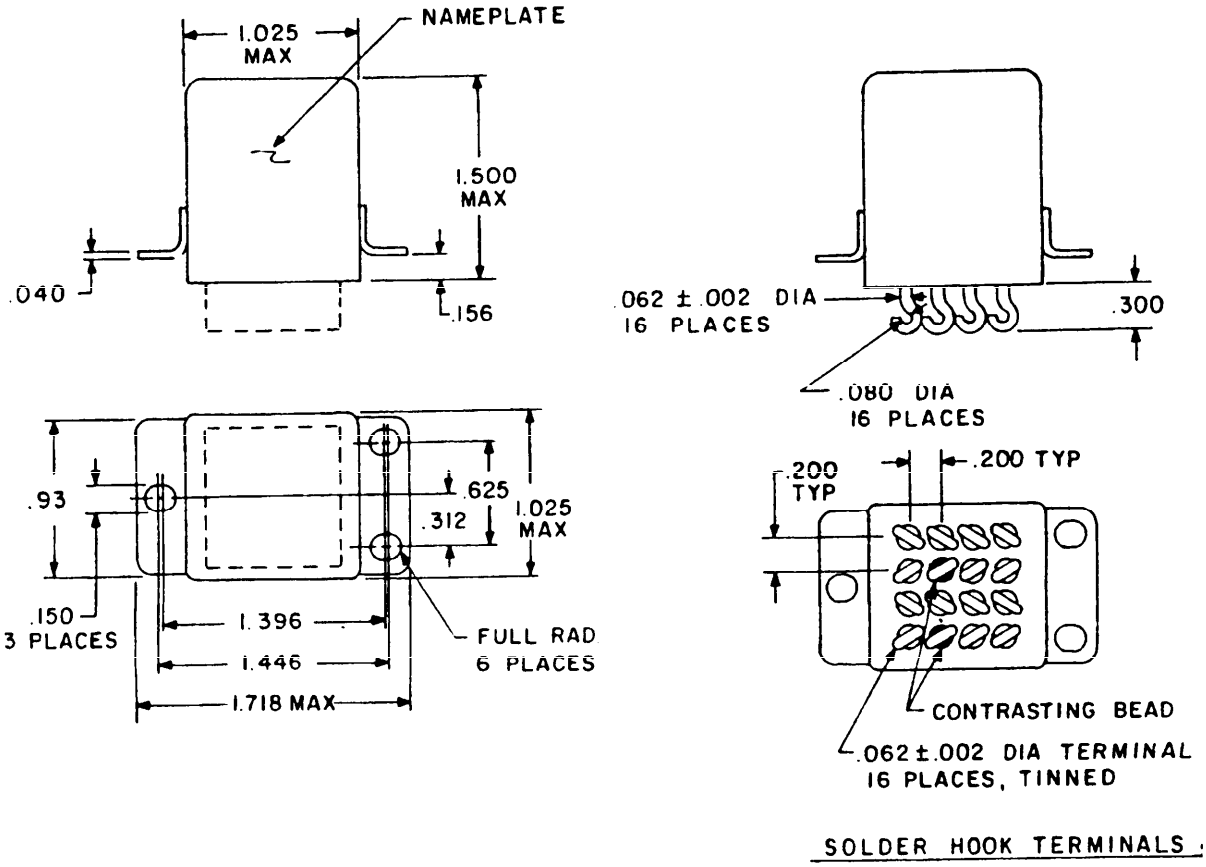
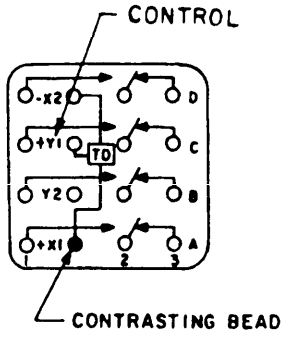


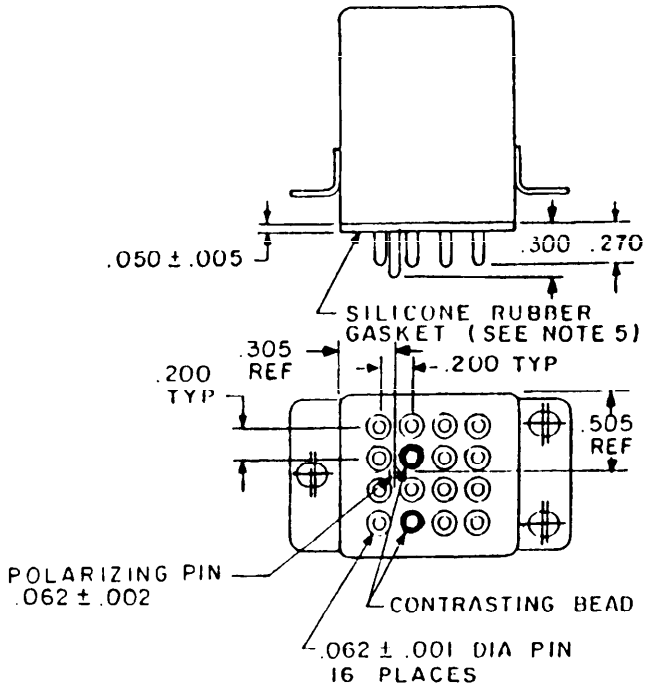
FIGURE 1. Outline dimensions and configuration of relay.

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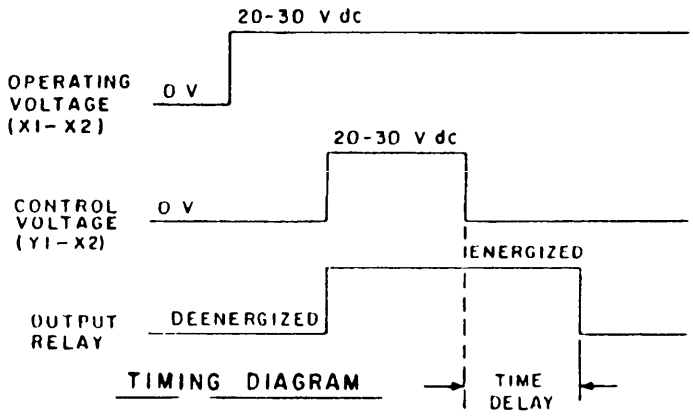
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CIRCUIT DIAGRAM (SEE NOTE 4)
TERMINAL VIEW



PLUG-IN TERMINALS
(SEE NOTE 6)



Inches	mm
.040	1.02
.050	1.27
.062	1.57
.080	2.03
.150	3.81
.156	3.96
.200	5.08
.270	6.86
.300	7.62
.312	7.92
.625	15.78
.937	23.80
1.025	26.04
1.396	36.73
1.445	36.73
1.500	37.10
1.718	43.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Terminal numbers shall not appear on the relay header and there shall be a legible circuit diagram on relay which identifies each terminal location specified.
5. Gasket material: Silicone rubber gasket AMS3332. Shore hardness 20 ±5.
6. Terminal plating: Gold in accordance with MIL-G-45204 type II, class 1. Underplating, nickel 50 to 150 microinches thick.
7. Terminal Y2 is not used.

FIGURE 1. Outline dimensions and configuration of relay - Continued.

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REQUIREMENTS:

OPERATING DATA:

Timing action: Delay-on release.

Time delay: Fixed; select from 0.1 to 600 seconds.

Timing accuracy 1/: ±10 percent of nominal value.

Recycle time: 50 milliseconds maximum.

Power interrupt 2/: 500 microseconds.

Operate current: 150 milliamperes maximum at 25°C.

Control voltage: 20 V dc to 30 V dc.

Control current: (Y1 - X2) 15 milliamperes maximum at 25°C.

INPUT DATA:

Input voltage 3/: 20 to 30 V dc.

Duty rating: Continuous.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

OUTPUT DATA:

Configuration: 4PDT.

Life: See table I.

TABLE I. Life load ratings (relay case grounded).

Type of load	Life (cycles)	Amperes 28 V dc	Amperes 115/200 V ac 400 Hz
Resistive	100,000	10	10
Inductive	20,000	8	8
Motor	100,000	4	4
Lamp	100,000	2	2
Low level	100,000	Applicable in accordance with MIL-R-6106	---

ELECTRICAL DATA:

Transients: +80 volts, in accordance with MIL-STD-704, figure 9, limit 1.

Spike:

Self-generated: ±50 volts maximum.

Spike transients: ±600 volts, 10 microseconds maximum.

Susceptibility: +80 volts maximum, -600 volts maximum.

1/ The accuracy requirement applies for any combination of operating temperature and voltage. Add ±10 milliseconds for timing less than 1 second.

2/ Transient and power loss specification are based on a maximum duty cycle of 1/50.

3/ Minimum high temperature test 21 V dc; minimum continuous current test 23.5 V dc.

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Electromagnetic interference 4/: In accordance with MIL-STD-461, class 1D.

Insulation resistance 5/: 1,000 megohms at 500 V dc at sea level, and 100 V dc at 80,000 feet between each pin and case.

Dielectric withstanding voltage 5/: 1,000 V rms at 60 Hz at sea level, and 350 V rms at 80,000 feet between case and pins connected together.

ENVIRONMENTAL DATA:

Ambient temperature:

Operating: -55°C to +125°C.

Nonoperating: -65°C to +125°C.

Vibration (sinusoidal): 30 G, 10 to 3,000 Hz.

Vibration (random): 0.4 G^2/Hz power spectral density, 50 to 2,000 Hz in accordance with MIL-STD-810.

Shock (specified pulse): 100 G, 6 ±1 milliseconds, 1/2 sine, 3 axes.

Acceleration: 15 G in any axis.

Seal: Hermetic.

PHYSICAL DATA:

Dimensions and configuration: See figure 1.

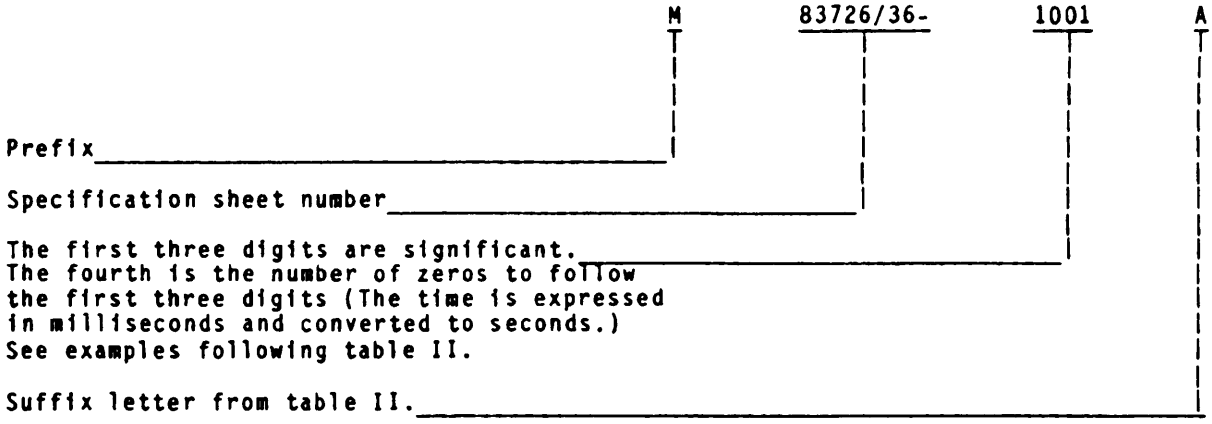
Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 3.0 ounces maximum.

Marking: See MIL-R-83726.

PART NUMBER 6/: Consists of the prefix, specification sheet number, a four-digit dash number (expressed in milliseconds), and a suffix letter from table II (indicating quality level) as follows:



4/ EMI test limits will not be exceeded during the timing interval or when continuously energized under steady-state conditions in accordance with EMI test of MIL-R-83726.

5/ Terminals X1, X2, and Y1 must be connected together during this test. Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and between all terminals and case.

6/ Time delay relays within the 0.1- to 600-second delay range are available.

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TABLE II. Suffix letters for quality levels.

Quality level indication (refer to MIL-R-83726)	P (plug-in)	S (solder-hook)
W	P	S
Y	A	B
X	C	D

Examples:

M83726/36-1001P: 1-second time delay, plug-in, W level.

M83726/36-9002B: 90-second time delay, solder hook, Y level.

M83726/36-6003C: 600-second time delay, plug-in, X level.

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F637-5)

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